Comprehensive Project Presentation Guide: CloudOptimizer Autopilot

Autonomous Cloud Cost & Performance Rightsizing Engine for AWS, Azure, and Google Cloud

Executive Summary

CloudOptimizer is a closed-loop, machine-learning-driven cloud management platform.

By fusing supervised time-series forecasting (LightGBM) with heuristic policy guardrails,

CloudOptimizer delivers **21.8% to 27% net verified cloud spend reduction** while

upholding strict **99.01% SLA compliance** with automated 1-click execution and instant

rollback capabilities.

1. High-Level Overview (The "Elevator Pitch")

> **What is CloudOptimizer in simple terms?**
>
> CloudOptimizer is an **AI-powered Autopilot for Cloud Infrastructure** (AWS, Microsoft Azure, and Google Cloud). In modern software companies, developers spin up virtual servers, forget about idle test machines running 24/7, or drastically oversize servers "just in case" to prevent crashes. This leads to massive wasted cloud spending.
>
> CloudOptimizer continuously ingests telemetry and billing reports, uses **Supervised**

> **Machine Learning** to forecast upcoming workloads 7 days into the future, identifies idle and over-provisioned resources, and safely recommends exact rightsizing actions that cut monthly cloud bills by **21.8% to 27%** with **zero downtime** and **99.01% SLA compliance**.

2. Why Traditional Tools Fail vs. What We Built

Traditional Cloud Management	CloudOptimizer Autopilot
Reactive autoscaling: Scales servers *after* CPU spikes, causing lag, buffer queuing, and sluggish user experience.	**Proactive ML demand forecasting:** Predicts traffic spikes up to 7 days ahead so servers are rightsized *before* traffic surges hit.
Assumption-based sizing: Guesses server load based purely on invoices and static thresholds.	**Data-driven monitoring:** Pairs billing invoices with true CPU, memory, storage, and network telemetry.
Dangerous automated shutdowns: Blindly turns off machines, risking production outages and SLA breaches.	**Closed-loop safety guardrails:** Enforces production protection rules, calculates prediction uncertainty bands, and provides **1-Click Undo rollback**.
Complex cloud jargon: Confusing charts with no clear advice or remediation steps.	**Explainable AI in Plain English:** Every suggestion includes an expandable **"Why?" Evidence Drawer** detailing metrics, forecasts, and policy reasons.

3. Technology Stack & Architecture

3.1 Frontend (Client-Side)

- **Language:** Vanilla JavaScript (ES6+), HTML5, CSS3.
- **Design Rationale:** Zero heavy framework dependencies (no React/Angular overhead) ensures sub-150ms load times, zero external attack surface, and instantaneous client-

side filtering and rightsizing simulations.

- **Styling:** Tailwind CSS using an accessible Warm Beige (`#fbf9f5`) and Sage Green (`#2e7d32`) palette.
- **Interactive Charts:** Chart.js 4.4 with custom uncertainty confidence band rendering.
- **Authentication & Storage:** Firebase SDK v10 (Google One-Tap OAuth2 and Firebase Cloud Storage).
- **Deployment:** Netlify Edge CDN with atomic Git CI/CD and client-side SPA routing.

3.2 Backend (API & ML Engine)

- **Language:** Python 3.12.
- **API Framework:** FastAPI with asynchronous endpoints, Pydantic schemas, and Swagger OpenAPI documentation.
- **Data Vectorization:** Pandas & NumPy for chunked telemetry parsing, currency normalization, and outlier detection.
- **Machine Learning:** LightGBM Regressor for demand forecasting with autoregressive lag features and moving averages.
- **Policy Engine:** YAML-based guardrails (`rules.yaml`) enforcing production protection, maintenance windows, and safe CPU limits.
- **Automated Testing:** 31 comprehensive unit and integration tests executed with Pytest.

4. The Training Dataset

The AI forecasting engine was calibrated using real-world enterprise cloud workloads:

4.1 Primary Dataset: Microsoft Azure Public VM Trace V1

- **Origin:** Microsoft Research Azure Public Dataset (from enterprise production compute clusters).
- **Scope:** 16 consecutive days (1,382,100 seconds) of raw production cloud telemetry.
- **Raw Scale:** 1,371,194 five-minute telemetry records processed through streaming pipelines.

- **Representative Cohort:** 300 virtual machines with continuous, unbroken timelines across 4 enterprise workload profiles:
    1. *Steady Workloads:* Consistent baseline utilization with modest variance.
    2. *Daily Peak Workloads:* Business applications with diurnal cycles (peak in afternoon, low at night).
    3. *Bursty Workloads:* Microservices with sharp query spikes.
    4. *Mostly Idle Workloads:* Zombie test servers with mean CPU < 10%.
- **Hourly Aggregation:** Streamed 5-minute telemetry resampled into hourly average CPU (`hourly_avg_cpu`) and peak CPU (`hourly_max_cpu`) (384 continuous hours per VM, 115,200 structured records).

4.2 Multi-Cloud Fleet Benchmark

Synthesized cross-cloud dataset covering 12 benchmark production and test resources across AWS, Azure, and GCP to validate cross-cloud normalization across 7 standardized dimensions.

5. Machine Learning Model Accuracy & Metrics

To prevent data leakage, the dataset was split chronologically:

- **Training Set (70%):** Hours 168 to 318 (45,300 samples)
- **Validation Set (15%):** Hours 319 to 350 (9,600 samples)
- **Unseen Test Set (15%):** Hours 351 to 383 (9,900 samples)

5.1 Model Comparison on Unseen Test Data

Model	MAE	RMSE	sMAPE	Overall Accuracy	Status
Naive (t-1 lag)	1.992%	3.014%	11.26%	88.74%	Baseline
Seasonal Naive (t-168 lag)	1.211%	1.904%	9.14%	90.86%	Passed
Moving Average (24-Hour)	4.620%	8.527%	16.03%	83.97%	Sub-optimal

Model	MAE	RMSE	sMAPE	Overall Accuracy	Status
Random Forest Regressor	0.993%	1.466%	7.25%	92.75%	High
LightGBM Regressor (Production)	**0.967%**	**1.449%**	**7.04%**	**92.96%**	**Best Performance**

5.2 Overfitting Audit

- **Training Accuracy:** 93.17% (MAE: 0.920%)
- **Validation Accuracy:** 92.99% (MAE: 0.974%)
- **Unseen Test Accuracy:** 92.96% (MAE: 0.967%)
- **Audit Conclusion:** The test accuracy closely tracks training accuracy within 0.21%, confirming zero overfitting and outstanding real-world generalization.

5.3 Financial & Operational Impact

- **Cloud Cost Reduction:** 26.96% net verified reduction in cloud compute spend.
- **Enterprise SLA Reliability:** 99.01% compliance (overload rate < 0.01%, only 1 hour over the 90% CPU limit out of 9,900 VM-hours).

6. The 7 Canonical Telemetry Dimensions

CloudOptimizer ingests files in CSV, Excel (`.xlsx`), or JSON format and cleans them into 7 standardized dimensions:

1. **Resource Identity:** ID, name, workload type, cloud provider (AWS, Azure, GCP), region, and current sizing tier.
2. **CPU Metrics:** Average CPU %, Peak CPU %, and multi-day trend patterns.
3. **Memory Metrics:** RAM utilization %, total allocated GB, and available capacity.
4. **Storage Metrics:** Total disk capacity GB, used GB, and attachment status (detects unattached disks).
5. **Network Metrics:** Inbound and outbound bandwidth throughput in MB.

6. **Billing Information:** Effective monthly cost, service cost, and currency.
7. **Usage History:** Uptime hours, daily runtime schedules, and history points.

7. How the Optimization Engine Works

1. **Ingestion & Normalization:** Strips currency noise, reconciles CPU scales (e.g., 0.15 vs 15%), and infers cloud providers.
2. **Classification:** Categorizes instances into *Idle* (< 5% CPU), *Over-Provisioned* (< 20% CPU), *Healthy* (40-70%), or *Hot* (> 85%).
3. **7-Day Demand Forecast:** LightGBM forecasts demand for the next 168 hours with upper and lower confidence bounds.
4. **Safety Policy Guardrails:** Verifies that post-downsizing peak CPU stays below safe limits (Careful: 65%, Balanced: 75%, Aggressive: 85%). Production termination requires explicit manual authorization.
5. **Execution & Rollback:** Generates Infrastructure-as-Code (Terraform plan and Kubernetes `kubectl` commands), runs post-change health verification, and logs actions to an immutable audit trail with 1-click Undo.

8. Complete Tour of Features Inside the App

- **Home (Dashboard):** Top KPI cards (Monthly Cost, Possible Savings, Items Needing Attention), plain-English summary, and multi-cloud telemetry banner.
- **Suggestions:** Live risk filtering (Low, Medium, High), environment filtering (Production vs. Test), instant text search, 1-click approval/rejection, and expandable "Why?" Evidence Drawers.
- **My Servers:** Complete inventory table with cloud badges, live search by server ID or type, provider filter, status filter, and 7-day forecast modal with uncertainty bands.
- **Try a Change (What-If Simulator):** Test downsizing or upsizing scenarios before making real-world modifications, complete with immediate safety verdict badges.
- **Apply & Undo:** Review approved changes, execute infrastructure updates, run automated health checks, and trigger 1-click instant rollback.

- **Settings:** Select policy presets (Careful, Balanced, Aggressive), adjust monthly budgets, configure FastAPI endpoints, and manage Firebase storage.
- **History:** Full chronological audit trail of all approved, rejected, and executed actions.

9. Presentation Q&A Cheat Sheet

Q1: "How is your model different from AWS Cost Explorer or Azure Advisor?"

> **Answer:** "AWS and Azure advisor tools work in proprietary silos—they only see their own cloud and usually give static, backward-looking suggestions based on average billing data. CloudOptimizer is **multi-cloud** (normalizing AWS, Azure, and GCP together) and uses **proactive machine learning** to forecast future workload demand 7 days ahead with uncertainty bands, preventing accidental outages from seasonal traffic spikes."

Q2: "What happens if the model makes a mistake and downsizes a critical server?"

> **Answer:** "We built a multi-layered safety system: First, our **Production Protection Policy** blocks aggressive downsizing on production servers unless > 75% headroom is guaranteed. Second, we provide an **Interactive What-If Simulator** to test the change beforehand. Third, changes generate **Terraform / kubectl** code with automated post-change health checks, and our **1-Click Undo button** instantly rolls back any applied change."

Q3: "Why did you choose LightGBM over deep learning (like LSTMs or Transformers)?"

> **Answer:** "Cloud telemetry tabular time-series data with strong cyclical patterns (hourly, daily, weekly lags) is where gradient-boosted decision trees excel. LightGBM achieved **92.96% accuracy** with an inference latency of just **2.4 milliseconds**, whereas neural networks require heavy GPU infrastructure and are prone to overfitting on sparse enterprise server logs."

Q4: "Where can anyone try the project right now?"

> **Answer:** "The application is live on the web at **cloud-optimizer.netlify.app** with open-source code on GitHub at **github.com/wsaman01u-tech/CLOUD_Optimizer**. Anyone can click 'View live demo' to explore the full interactive Azure trace sample, or sign in with their Google account to upload and optimize their own multi-cloud infrastructure."